

1 (a) Complete the truth table for the logic expression:

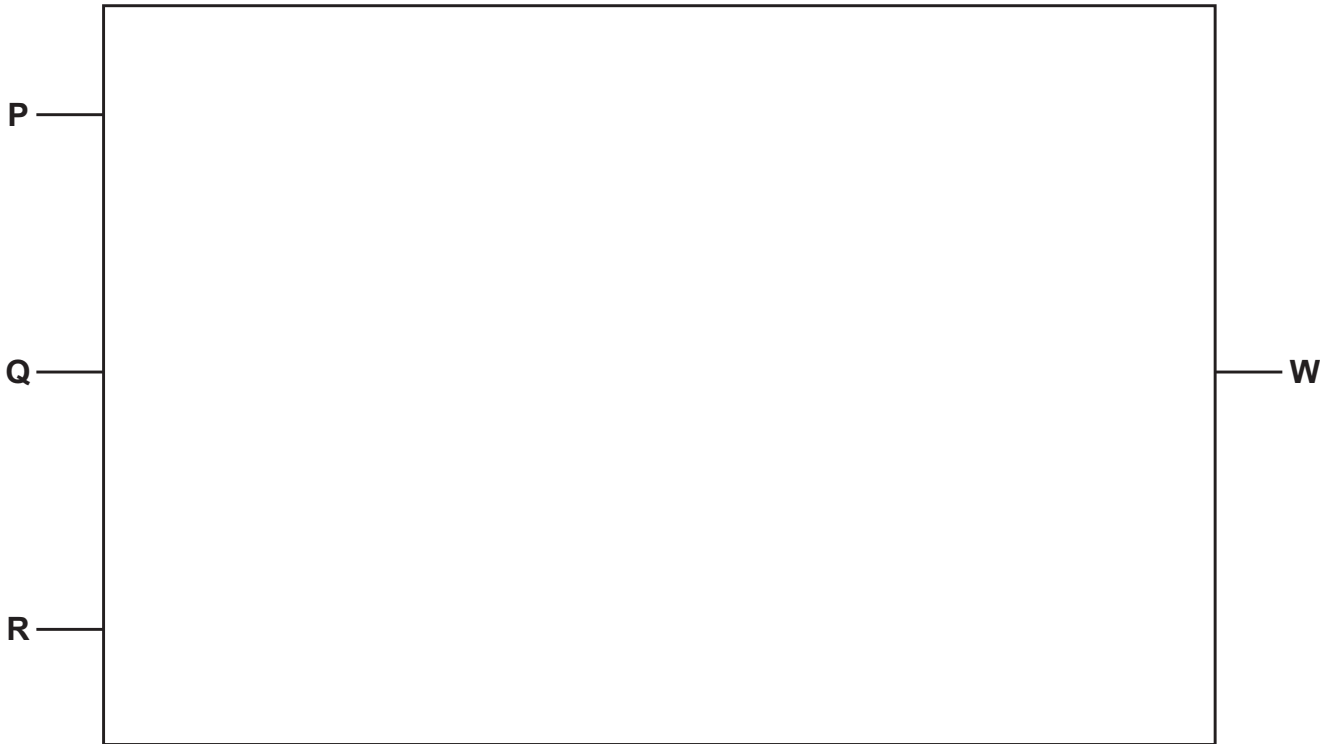
$$X = (A \text{ XOR } B) \text{ NAND } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(b) Draw the logic circuit for the logic expression:

$$W = P \text{ NAND } ((Q \text{ OR NOT } R) \text{ XOR } (P \text{ XOR } Q))$$



[2]